

1. Prepaid Charging for Voice Call

We will emulate voice calls and its real-time charging. We will have three applications:

- First application(Mobile Phone) will act as mobile phone which will capture the voice from microphone and send it via UDP protocol to second application (MSC). This first application (Mobile Phone) when started will start send UDP traffic and will print every one minute on screen/stdout the elapsed minutes from start the mobile application
- Second Application (MSC) which will receive traffic from UDP protocol and play the voice data at PC Speaker. And the MSC will deduct from User Balance every one minute 1 L.E per minute. And at the end of the voice call the application will generate CDR with call details.
- The Third application will be web interface (HTML, CSS, Servlet) to make CRUD operation on DB for table Users (ID, MSISDN, Balance)

The first application will start as following command line:

```
Java mobile <MSISDN>
```

Such as :

```
Java mobile 01223456789
```

And its output will be such as:

```
Starting voice call as MSISDN 01223456789
Capturing Voice from Microphone and send via UDP.....
1 minutes elapsed
2 minutes elapsed
3 minutes elapsed
4 minutes elapsed
5 minutes elapsed
```

The Second application will start with the following command line:

Such as :

```
Java msc
```

And its output will be such as:

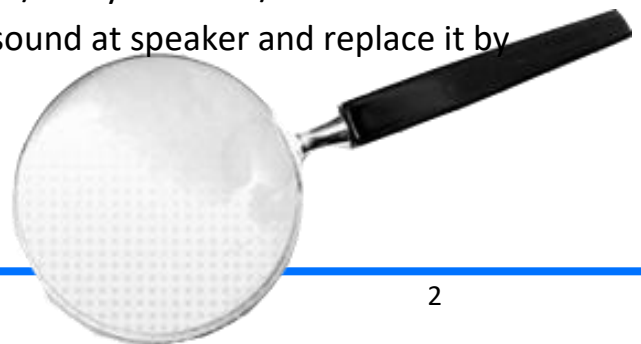
```
Waiting for voice call Signaling start message via TCP
Accept Voice call start signaling message from MSISDN 01223456789
Capturing UDP traffic and play via speaker .....
Call End after receiving end call signaling message
```

Generating CDR line: 01223456789, 2025-03-01T15:26:47.398253 , 2025-03-01T15:24:47.398253 ,4,Normal call Clearing,20,1001

- When the first application (Mobile Phone) start will get the MSISDN via main(Arguments) and send to second application via TCP Protocol via 'Start Call <MSISDN>' Signaling message to identify itself to MSC application and when the first application (Mobile Phone) shutdown (addShutdownHook/....), you send to second application (MSC) that call us end 'End Call' to stop charging
- The Second application (MSC) will start receiving UDP Traffic and play it via Speaker after received via TCP Connection 'Start Call' signaling message
- The Second application (MSC) will deduct balance from this MSISDN which received at 'Start Call' signaling message every one minute fixed amount 1 L.E per minute.
- The Second application (MSC) will stop charging and stop play audio traffic and stop received from UDP after receiving 'End Call' signaling message
- The Second application (MSC) will calculate the call Cost and append CDR to file /tmp/calls.cdr
- Generate CDR file at end of call contain following info:
 - MSISDN
 - Start Time
 - End Time
 - Duration
 - Call result(normal call clearing, user not found on DB)
 - Call cost
 - Balance after call end

Bonus Requirements:

- Replace UDP with RTP Protocol
- Make the CDR file to rotate every one hour (you can use log4J Library with rotate features) such as:
/tmp/calls_CDR_2025_03_02_10.cdr
- Save the voice data of call to file with this format
/tmp/voice_call_msisdn_01223355_date_2026_06_01_Time_10_20_30.wav/mp3
- Make MSC application to support concurrent/many and calls/Mobiles at same time, and in this case, you will disable play sound at speaker and replace it by saving voice data to file.



2. SMS Client via SMPP Protocol

Make new copy (GitHub Fork) of Twilio SMS Client and replace Twilio API by SMPP Protocol and use SMPP simulator tool to simulate Twilio backend.

Make new Field at SMS Submit form called Validity period. The default value should be 5 minutes.

Add new feature of 'Forward and Store' at SMS Client project by using Scheduler in case of SMS failed to deliver as first attempt, you will add SMS to DB so the scheduler Task/app/process will try to deliver SMS at fixed intervals. And SMS/row will be deleted from DB once the Delivery is successfully.

